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INFO 4220 – Networks II: Market Design
Spring 2024 – Prelim 2

Problem 1 (12 points)

[This might be familiar to those who did PS3] In a world where cities frequently face challenges ranging from alien invasions to supervillain schemes, a council of superheroes has been formed to protect humanity. There are three cities (Silicon City, Greenburg, and Metroville), and the council includes three renowned heroes, each with their own unique powers and city preferences:

Captain Fantastic (CF)	Ms. Mystic (MM)	The Speedster (TS)
Silicon City	Silicon City	Metroville
Metroville	Greenburg	Silicon City
Greenburg	Metroville	Greenburg

Given these preferences and the unique needs of each city, the council is trying to find a fair method to determine who goes where, ensuring each city gets the hero it needs while considering the heroes' wishes.

a) Find the probabilistic assignment that results from using a random serial dictatorship in this problem. (5 points)

Solution:

There are six possible orders of priority. Below, the table shows, for each order, the corresponding assignment.

Order	Captain Fantastic (CF)	Ms. Mystic (MM)	The Speedster (TS)
CF>MM>TS	Silicon City	Greenburg	Metroville
CF>TS>MM	Silicon City	Greenburg	Metroville
MM>TS>CF	Greenburg	Silicon City	Metroville
MM>CF>TS	Metroville	Silicon City	Greenburg
TS>CF>MM	Silicon City	Greenburg	Metroville
TS>MM>CF	Greenburg	Silicon City	Metroville

(3 points for the matrix with the six possible assignments; -0.5 for each incorrect assignment)

By calculating the frequencies of this assignments, we end up with the following random assignment:

	Captain Fantastic (CF)	Ms. Mystic (MM)	The Speedster (TS)
Silicon City	1/2	1/2	0
Greenburg	1/3	1/2	1/6
Metroville	1/6	0	5/6

(2 points for the correct random assignment; 1 point if they correctly sum over the frequencies BUT their matrix above is incorrect).

b) If instead of using a random serial dictatorship you use the top trading cycle with random endowments. Would the assignment be different from the one you found in (a)? Why? (2 points)

Solution

In random assignment problems, the random serial dictatorship and the top trading cycle algorithm are equivalent. Therefore, the solution would not change.

(1.5 points if they said that it wouldn't change. 0.5 points if they motivate why).

c) Find a random assignment that fulfills the following two conditions: (1) assigns both Captain Fantastic and Ms. Mystic a 50% probability of getting Silicon City, and (2) cannot be stochastically dominated by any other assignment for Captain Fantastic and Ms. Mystic (it can be stochastically dominated by other assignments for The Speedster). (5 points)

Solution

In order to get a random assignment that is not stochastically dominated for CF and MM, we need both to maximize the probability of getting their second top choice.

Claim 1: given that the random assignment we are going to find will assign both 50% probability of getting their top choice (Silicon City), the probability they will get their second choice **must be exactly 50%**.

Proof: by contradiction, assume that the probability of getting their second choice is less than 50%. Since their second top choice is not conflicting (they desire different things, CF desires Metroville, MM prefers Greenburg), having less than 50% probability of getting that choice would be stochastically dominated by another assignment that gives them higher probability of obtaining their second top choice. Also, any assignment that gives them probability higher than 50% of obtaining their second choice would be unfeasible. Therefore, this assignment must give both of them exactly 50% of obtaining their second choice. We *deduce* that TS will have 50% probability of getting their second (or third) choices as follows:

	Silicon City	**Greenburg**	**Metroville**
CF	1/2	0	1/2
MM	1/2	1/2	0
TS	0	1/2	1/2

There is no other assignment that will stochastically dominate the assignment above for CF and MM, and that assigns probability 50% of getting Silicon City to each of the two. Proof about stochastic dominance for both CF and MM is provided above.

(3 points for the correct matrix. 2 points for explanation of why it is stochastic dominant for CF and MM. Prove does not need to be as comprehensive as above. As long as they recognize and explain that they have to give 50% to the second choice, it is OK.]

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Problem 2 (16 points)

Consider a school choice problem as below. The students' true preferences and the weak priorities at these schools are given in the following table.

Alice	**Bob**	**Carol**	**Denis**
Berkeley	Berkeley	Yale	Yale
Yale	Cornell	Cornell	Cornell
Cornell	Yale	Berkeley	Berkeley

Cornell [cap = 1]	**Yale [cap = 1]**	**Berkeley [cap = 2]**
[Bob, Carol]	Alice	[Bob, Denis]
Alice	[Carol, Denis]	Carol
Denis	Bob	Alice

To break down ties, the schools use a single tie breaking rule. For parts (a) – (d) consider that a lottery was done and each of the students got the following priority when there is a tie: **Bob> Denis>Carol**. For parts (e) and (f) consider all possible orders of students.

a. Use the deferred acceptance algorithm with student proposing to find an assignment. (2 points)

Solution:

Step 1: Alice -> Berkeley, Bob -> Berkeley, ~~Carol -> Yale~~, Denis -> Yale

Step 2: Carol -> Cornell.

Assignment: $\mu(Alice) = \text{Berkeley}, \mu(Bob) = \text{Berkeley}, \mu(Carol) = \text{Cornell}, \mu(Denis) = \text{Yale}$

1pt for the correct process (0.5pt for each step x2), 1pt for the correct result.

b. Explicitly evaluate if the assignment you found in (a) is stable (i.e. don't use the result that the DA algorithm produces stable assignments). (2 points)

Solution:

0.5pt for stating the assignment is stable. 0.5pt for each condition (0.5pt x3) The students need to justify why each condition is met.

The assignment is stable. We need 3 conditions:

- No individual irrationality: It's because no one is matched to an unacceptable school.
- No wastefulness: because all school are full.
- No justified envy: Alice, Bob and Denis don't envy anyone. Carol envies Denis, but it's not justified because they both have same priority at Yale.

c. Explicitly evaluate if the assignment you found in (a) is efficient. (2 points)
Solution:
The assignment is efficient. Only Carol is not assigned to her favorite school, and to get it to her we need to take it away from Denis, who will be worse off.
Alternative answer: Alice, Bob and Denis have their most preferable assignment, so Carol cannot swap with anyone to get a more preferable assignment while not making anyone else worse off.

1pt stating the assignment is efficient, 1pt for sufficient explanation.

d. Consider another assignment where $\mu(Alice) = \text{Berkeley}, \mu(Bob) = \text{Cornell}, \mu(Carol) = \text{Yale}, \mu(Denis) = \text{Berkeley}$. Your friend says that Bob justifiably envies both Alice and Denis. Do you agree with your friend or not? Clearly explain your answer. (2 points)

Solution: I don't agree.
The assignment is unstable, but Bob only justified envies Alice since he has a strictly higher priority at Berkeley than Alice, while Bob does not justify envy Denis for Yale **since they have the same priority**.

1pt for stating not agree/friend's statement is incorrect. 1pt for explaining Bob does not justified envy Denis because they have the same priority at Yale.

e. How many stable assignments are possible in this problem assuming all students submit their true preferences? Find them. (Hint: Look at the preferences and priorities and think carefully before trying to solve this problem.) (4 points)

Solution: There are 2 stable assignments. Alice and Bob always get Berkeley, Carol and Denis alternate.

One assignment is $\mu(Alice) = \text{Berkeley}, \mu(Bob) = \text{Berkeley}, \mu(Carol) = \text{Cornell}, \mu(Denis) = \text{Yale}$.
The other assignment is: $\mu'(Alice) = \text{Berkeley}, \mu'(Bob) = \text{Berkeley}, \mu'(Carol) = \text{Yale}, \mu'(Denis) = \text{Cornell}$.

1pt for stating there are 2 stable assignments. 2pts for each stable assignment (2pts x2). Deduct points if listed other assignment that are unstable. Cap at 4pts.

f. If we use Deferred Acceptance followed by Improvement Cycles, can Denis misrepresent his preferences to obtain a better assignment? (Assume all other students submit their true preferences.) (4 points)

Solution: Yes, Denis could mispresent himself to obtain a better outcome. **(1pt)**

Explanation (3pts in total): Denis could put Berkeley first: [Berkeley, Yale, Cornell] or [Berkeley, Cornell, Yale]. **(1pt for showing how Denis can misrepresent his preference, saying Denis could put Berkeley first is enough)**
The result with DA will always be:

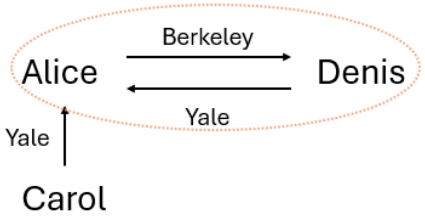
Step 1: ~~Alice -> Berkeley~~, Bob -> Berkeley, Carol -> Yale, Denis -> Berkeley

Step 2: Alice -> Yale (~~Carol -> Yale~~)

Step 3: Carol -> Cornell

$\mu''(Alice) = \text{Yale}, \mu''(Bob) = \text{Berkeley}, \mu''(Carol) = \text{Cornell}, \mu''(Denis) = \text{Berkeley}.$
(1pt for showing the correct DA result, no need to show the steps to get full credit. Deduct points if steps/explanation is incorrect.)

Now Denis can trade Berkeley for Yale with Alice using IC, and he could always get his favorite school: $\mu(Alice) = \text{Berkeley}, \mu(Bob) = \text{Berkeley}, \mu(Carol) = \text{Cornell}, \mu(Denis) = \text{Yale}.$



With this misrepresentation, Denis always gets his favorite school.

(1pt for showing IC or Denis can trade to get a better outcome. No need to draw the IC graph, but need to clearly explain how Denis should trade if not provides graph. Deduct points if steps/explanation is incorrect.)

Alternative Explanation (3 points in total): Denis could also put Berkeley second: [Yale, Berkeley, Cornell]. **(1pt for showing how Denis can misrepresent his preference, saying Denis could put Berkeley second is enough)**

- If Carol<Denis for Yale, the assignment from DA is $\mu(Alice) = \text{Berkeley}, \mu(Bob) = \text{Berkeley}, \mu(Carol) = \text{Cornell}, \mu(Denis) = \text{Yale}$, the same as revealing true preference and Denis gets his favorite school. No need to run IC (no IC can be run).
- If Carol>Denis for Yale, the assignment from DA is $\mu''(Alice) = \text{Yale}, \mu''(Bob) = \text{Berkeley}, \mu''(Carol) = \text{Cornell}, \mu''(Denis) = \text{Berkeley}.$
    - With IC, μ''can be improved to $\mu(Alice) = \text{Berkeley}, \mu(Bob) = \text{Berkeley}, \mu(Carol) = \text{Cornell}, \mu(Denis) = \text{Yale}.$

With this misrepresentation, Denis can always get his favorite school.

(1pt for discussing results from DA in different conditions, 1pt for showing IC or Denis can trade to get a better outcome. No need to draw the IC graph, but need to clearly explain how Denis should trade if not provides the graph. Deduct points if steps/explanation is incorrect.)

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Problem 3 (12 points)

There are 8 patient-donor pairs for a kidney exchange situation: (Javier, Amelia), (Andrea, Jose), (Helena, Diego), (Jason, Peter), (Lee, Charles), (Maria, Miguel), (Joy, Fiona), (Robert, Pamela). In the following table, you will find their blood types and gender:

Donors	
Name	Blood type
Amelia	A
Jose	AB
Diego	A
Peter	AB
Charles	B
Miguel	A
Fiona	O
Pamela	O

Patients	
Name	Blood type
Javier	O
Andrea	O
Helena	B
Jason	O
Lee	O
Maria	B
Joy	AB
Robert	A

Recall that blood type compatibility is as follows:

- O patients can only receive from O donors.
- A patients can receive from A and O donors.
- B patients can receive from B and O donors.
- AB patients can receive from any donor.

The preferences of patients over organs is as follows:

Javier	**Andrea**	**Helena**	**Jason**	**Lee**	**Maria**	**Joy**	**Robert**
Fiona	Pamela	Pamela	Fiona	Pamela	Pamela	Peter	Amelia
Pamela	Fiona	Fiona	Pamela	Fiona	Fiona	Pamela	Diego
w	W	Charles	w	w	Charles	Miguel	Fiona
		w			w	Jose	Miguel
						Fiona	Pamela
						Diego	w
						Charles	
						Amelia	
						w	

a) If no organ exchanges are allowed, how many patients would be able to receive an organ in this problem. (2 points)

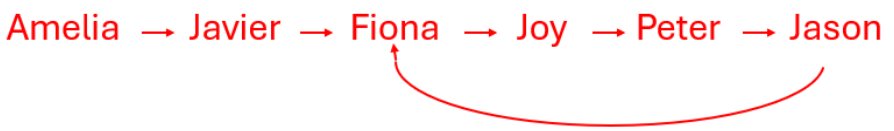
Only 2: Fiona – Joy and Pamela – Robert.

1 point each.

b) Apply the top trading cycles and chains algorithm using the following chain selection rule: Choose the longest chain and remove from the problem the patients and kidneys in that chain. If there are multiple chains of the same length, choose the chain based on the first letter of the name of the first donor in the chain in alphabetical order (i.e. if the two longest chains are of equal length, and one starts with Alice and the other with Bob, choose the chain that starts with Alice). (6 points)

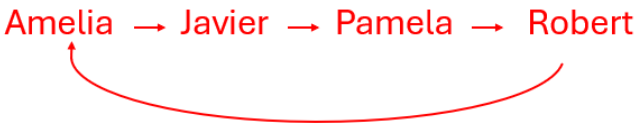
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First step:



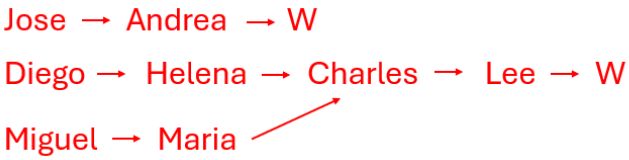
Assignments:
Peter – Joy
Fiona – Jason

Second step (In theory the diagram should show all patients, but we are going to say it is OK to not show the other patients as they will be chains, and we will only use the cycle from this step):



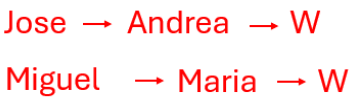
Assignments:
Amelia - Robert
Pamela - Javier

Third step:



Assignments: Chain selection rule says we must choose the chain starting with Diego
Lee joins the waiting list
Charles – Helena
Diego's kidney goes to someone in W

Fourth step:



Assignments:
Andrea and Maria are added to the waiting list
Jose's and Miguel's kidney goes to someone in the waiting list

c) Based on your response to part (b). How many patients in this problem get a kidney, how many join the waiting list, and how many donors donate their kidney to someone in the waiting list? (2 points)

5 patients get a kidney **(0.6 points)**

3 patients join the waiting list **(0.7 points)**

3 donors donate to the waiting list **(0.7 points)**

d) Fiona and Pamela are universal donors. Why should they participate in this kidney exchange if they could directly donate to their intended patient? (2 points)

Because by joining the kidney exchange their patients get a more preferred kidney **(1 point)** and 3 other patients get a kidney **(1 point)**.